

Statistics for Data Science - 1

Week 12 Graded Assignment

Continuous random variables

1. Let a random variable is uniformly distributed over $[a, b]$ with expectation and variance e and v respectively. Find the value of ab .

Answer: $e^2 - 3v$

Solution:

Given random variable is uniformly distributed from $[a, b]$.

$$E[X] = \frac{b+a}{2} = e \Rightarrow (b+a)^2 = e^2 \times 4 = 4e^2$$

$$Var(X) = \frac{(b-a)^2}{12} = v \Rightarrow (b-a)^2 = 12v$$

$$(b+a)^2 - (b-a)^2 = 4ab = 4e^2 - 12v$$

$$\Rightarrow ab = e^2 - 3v$$

For example: e=6 and v=4

Given random variable is uniformly distributed from $[a, b]$.

$$E[X] = \frac{b+a}{2} = 6 \Rightarrow (b+a)^2 = 6^2 \times 4 = 144$$

$$Var(X) = \frac{(b-a)^2}{12} = 4 \Rightarrow (b-a)^2 = 48$$

$$(b+a)^2 - (b-a)^2 = 4ab = 144 - 48 = 96$$

$$\Rightarrow ab = 24$$

2. Probability density function of a random variable X is given by

$$f(x) = \begin{cases} kx^2 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(m < X < n)$?

(Hint: $\int_a^b cx^2 dx = c \frac{b^3 - a^3}{3}$ where c is a constant)

(a). $n^2 - m^2$

(b). $\frac{n^3 - m^3}{9}$

(c). $\frac{n^2 - m^2}{9}$

(d). $n^3 - m^3$

Answer: d

Solution:

We know that $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\Rightarrow \int_0^1 kx^2 dx = 1$$

$$\Rightarrow k \frac{x^3}{3} \Big|_0^1 = 1$$

$$\Rightarrow k \frac{1}{3} = 1 \Rightarrow k = 3$$

We know that $P(m \leq X \leq n) = \int_m^n f(x)dx$

$$\Rightarrow P(m \leq X \leq n) = \int_m^n 3x^2 dx$$

$$\Rightarrow P(m \leq X \leq n) = 3 \frac{x^3}{3} \Big|_m^n$$

$$\Rightarrow P(m \leq X \leq n) = n^3 - m^3$$

For example: m=0.1 and n=0.7

We know that $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\Rightarrow \int_0^1 kx^2 dx = 1$$

$$\Rightarrow k \frac{x^3}{3} \Big|_0^1 = 1$$

$$\Rightarrow k \frac{1}{3} = 1 \Rightarrow k = 3$$

We know that $P(a \leq X \leq b) = \int_a^b f(x)dx$

$$\Rightarrow P(0.1 \leq X \leq 0.7) = \int_{0.1}^{0.7} 3x^2 dx$$

$$\Rightarrow P(0.1 \leq X \leq 0.7) = 3 \frac{x^3}{3} \Big|_{0.1}^{0.7}$$

$$\Rightarrow P(0.1 \leq X \leq 0.7) = 0.7^3 - 0.1^3 = 0.343 - 0.001 = 0.342$$

3. The amount of milk produced every day by a dairy is uniformly distributed between 100 litres and 120 litres. What is the probability that the amount of milk produced is more than m litres given that on that day, more than n litres of milk was produced?(Enter the answer correct to 2 decimal place)

Answer: $\frac{120 - m}{120 - n}$

Solution:

Given that milk produced is uniform between 100 and 120 litres.

Given that on a given day milk produced is more than n litres.

Let X denote the quantity of milk produced.

$$P(X > m | X > n) = \frac{1 - P(X \leq m)}{1 - P(X \leq n)} = \frac{1 - \left(\frac{m-100}{120-100}\right)}{1 - \left(\frac{n-100}{120-100}\right)} = \frac{120 - m}{120 - n}$$

For Example: m=115 and n=110

Given that milk produced is uniform between 100 and 120 litres.

Given that on a given day milk produced is more than 110 litres.

Let X denote the quantity of milk produced.

$$P(X > 115 | X > 110) = \frac{120-115}{120-110} = \frac{5}{10} = 0.5$$

4. The time that Jan shatabdi express will reach the Delhi station is uniformly distributed between x p.m. and y p.m. What is the probability that the train reaches Delhi exactly at z p.m?

Answer: 0

Solution:

The probability that train reaches exactly at z PM is $\int_z^z f(x)dx = 0$

Since the area under the curve at a particular instant of x value is zero.

For example: $x=12$, $y=5$ and $z=2$

The probability that train reaches exactly at 2 PM is $\int_2^2 f(x)dx = 0$

Since the area under the curve at a particular instant of x value is zero.

5. The length of time for one person to be served at a restaurant is exponentially distributed, with an expected time of t minutes. If 5 persons arrived at the restaurant, then what is the probability that at least 2 of them will be served in more than p minutes? (Assume that each person is served independently)

(a) $1 - \left(1 - e^{-\left(\frac{p}{t}\right)}\right)^5 - 5 \cdot e^{-\left(\frac{p}{t}\right)} \left(1 - e^{-\left(\frac{p}{t}\right)}\right)^4$

(b) $1 - \left(e^{-\left(\frac{p}{t}\right)}\right)^5 - \left[5 \left(e^{-\left(\frac{p}{t}\right)}\right)^4 \left(1 - e^{-\left(\frac{p}{t}\right)}\right)\right]$

(c) $1 - \left(1 - e^{-pt}\right)^5 - \left[5 \cdot e^{-pt} \left(1 - e^{-pt}\right)^4\right]$

(d) $1 - \left(e^{-pt}\right)^5 - \left[5 \cdot e^{-4pt} \left(1 - e^{-pt}\right)\right]$

Answer: a

Solution:

Given expected time $\frac{1}{\lambda} = t$ minutes.

$$\Rightarrow \lambda = \frac{1}{t}$$

$$P(X > p) = e^{-p\lambda} = e^{-\frac{p}{t}}$$

$$\Rightarrow P(X \leq p) = 1 - e^{-\frac{p}{t}}$$

Let i be the event that i among 5 of them are served in more than p minutes.

$$P(i) = {}^5C_i (P(X > p))^i (P(X \leq p))^{5-i}$$

$$P(i) = {}^5C_i \left(e^{-\frac{p}{t}}\right)^i \left(1 - e^{-\frac{p}{t}}\right)^{5-i}$$

$$P(i \geq 2) = 1 - P(i = 1) - P(i = 0)$$

$$P(i \geq 2) = 1 - {}^5C_1(e^{-\frac{p}{t}})^1(1 - e^{-\frac{p}{t}})^4 - {}^5C_0(e^{-\frac{p}{t}})^0(1 - e^{-\frac{p}{t}})^5$$

$$P(i \geq 2) = 1 - 5(e^{-\frac{p}{t}})(1 - e^{-\frac{p}{t}})^4 - (1 - e^{-\frac{p}{t}})^5$$

Hence, option (a) is correct.

For Example: t=5 and p=8

Given expected time $\frac{1}{\lambda} = 5$ minutes.

$$\Rightarrow \lambda = \frac{1}{5}$$

$$P(X > 8) = e^{-8\lambda} = e^{-\frac{8}{5}}$$

$$\Rightarrow P(X \leq 8) = 1 - e^{-\frac{8}{5}}$$

Let i be the event that i among 5 of them are served in more than 8 minutes.

$$P(i) = {}^5C_i(P(X > 8))^i(P(X \leq 8))^{5-i}$$

$$P(i) = {}^5C_i(e^{-\frac{8}{5}})^i(1 - e^{-\frac{8}{5}})^{5-i}$$

$$P(i \geq 2) = 1 - P(i = 0) - P(i = 1)$$

$$P(i \geq 2) = 1 - {}^5C_0(e^{-\frac{8}{5}})^0(1 - e^{-\frac{8}{5}})^5 - {}^5C_1(e^{-\frac{8}{5}})^1(1 - e^{-\frac{8}{5}})^4$$

$$P(i \geq 2) = 1 - (1 - e^{-\frac{8}{5}})^5 - 5(e^{-\frac{8}{5}})(1 - e^{-\frac{8}{5}})^4$$

6. Probability density function of a random variable X is given by

$$f(x) = \begin{cases} 0.2x & \text{if } 0 \leq x \leq 1 \\ 0.2 & \text{if } 1 \leq x \leq 2 \\ 0.2x - 0.2 & \text{if } 2 \leq x \leq 3 \\ 0.4 & \text{if } 3 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that $0 < X < 2.5$. (Answer upto three decimal points)

Hint: Try to find the area under pdf using integration, otherwise draw the graph of pdf of X

Answer: 0.425 accepted range: [0.421, 0.429]

Solution:

$$P(0 < X < 2.5) = \int_0^1 0.2x dx + \int_1^2 0.2 dx + \int_2^{2.5} (0.2x - 0.2) dx$$

$$P(0 < X < 2.5) = 0.1x^2|_0^1 + 0.2x|_1^2 + (0.1x^2|_2^{2.5} - 0.2x|_2^{2.5})$$

$$P(0 < X < 2.5) = 0.1 + (0.2 * 2 - 0.2) + (0.1 * 2.5^2 - 0.1 * 2^2 - 0.2 * 2.5 + 0.2 * 2)$$

$$P(0 < X < 2.5) = 0.425$$

7. In a certain exhibition, the time for the next visitor to come is exponentially distributed with a standard deviation of t minutes. What will be the expected time (in minutes) for two visitors to arrive if one comes after the other? (Note that the arrival of a visitor is independent of the arrival of the previous visitor)

- (a). $\frac{1}{t}$
 (b). $2t$
 (c). t
 (d). $\frac{2}{t}$

Answer: b

Solution:

Given standard deviation = t minutes.

$$\Rightarrow \text{Variance} = t^2$$

$$\text{Variance} = \frac{1}{\lambda^2} = t^2$$

$$1/\lambda = t \Rightarrow E[X] = 1/\lambda = t$$

Expected time for two visitors = $2 * E[X] = 2t$ minutes, since both are independent of other visitor.

For Example: $t=10$

Given standard deviation = 10 minutes.

$$\Rightarrow \text{Variance} = 10^2 = 100$$

$$\text{Variance} = \frac{1}{\lambda^2} = 100$$

$$1/\lambda = 10 \Rightarrow E[X] = 1/\lambda = 10$$

Expected time for two visitors = $2 * E[X] = 20$ minutes, since both are independent of other visitor.

8. The total duration (in minutes) of a badminton match in the Premier Badminton League (PBL) is uniformly distributed between, $[a, b]$ with variance 12 square minutes. The probability that a match will last at most m minutes is $\frac{1}{p}$. Find the expected time duration (in minutes) of a badminton match. (Enter the answer correct to 1 decimal place)

Solution:

Given Variance = 12 square minutes.

$$\Rightarrow \frac{(b-a)^2}{12} = 12$$

$$\Rightarrow (b-a)^2 = 12^2 \Rightarrow b-a = 12$$

$$\text{Given } P(X \leq m) = \frac{1}{p}$$

$$\Rightarrow P(X \leq m) = \frac{m-a}{b-a} = \frac{1}{p}$$

$$\Rightarrow m - a = \frac{b - a}{p}$$

$$a = m - \left(\frac{b - a}{p}\right) \Rightarrow b = m - \left(\frac{b - a}{p}\right) + 12$$

$$E[X] = \frac{b + a}{2}$$

For example: m=42 and p=6

Given Variance = 12 square minutes.

$$\Rightarrow \frac{(b - a)^2}{12} = 12$$

$$\Rightarrow (b - a)^2 = 12^2 \Rightarrow b - a = 12$$

Given $P(X \leq 42) = \frac{1}{6}$

$$\Rightarrow P(X \leq 42) = \frac{42 - a}{b - a} = 1/6$$

$$\Rightarrow 42 - a = 12/6 = 2$$

$$a = 40 \Rightarrow b = 52$$

$$E[X] = \frac{b + a}{2} = \frac{40 + 52}{2} = 46.$$

9. The probability density function of the time X (in minutes) between calls at the customer care is given by

$$f(x) = \begin{cases} \frac{1}{m} \cdot e^{-\frac{x}{m}} & \text{if } 0 \leq x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that time between calls exceeds the mean time.

- (a) $1 - \frac{1}{e}$
- (b) $1 - e$
- (c) $\frac{1}{e}$
- (d) e

Answer: c

Solution:

Mean time is $\frac{1}{\lambda} = \frac{1}{\frac{1}{m}} = m$

$$P(X \geq t) = e^{-\lambda t}$$

$$P(X \geq \frac{1}{\lambda}) = e^{-\lambda * \frac{1}{\lambda}} = e^{-1} = \frac{1}{e}$$

For example: m=4

Mean time is $\frac{1}{\lambda} = \frac{1}{\frac{1}{4}} = 4$

$$P(X \geq t) = e^{-\lambda t}$$

$$P(X \geq \frac{1}{\lambda}) = e^{-\lambda * \frac{1}{\lambda}} = e^{-1} = \frac{1}{e}$$

10. The lifetime of a light bulb is exponentially distributed with a mean life of n months. If there are $p\%$ chances that a light bulb will last at most t months, then what is the value of t ?

(a). $n \times \ln \left(\frac{100 - p}{100} \right)$

(b). $n \times \ln \left(\frac{100}{100 - p} \right)$

(c). $\frac{1}{n} \ln \left(\frac{100 - p}{100} \right)$

(d). $\frac{1}{n} \ln \left(\frac{100}{100 - p} \right)$

Answer: b

Solution:

Given mean of exponential random variable (life of light bulb) = n months.

$$\Rightarrow \frac{1}{\lambda} = n, \Rightarrow \lambda = \frac{1}{n}.$$

$$\text{Given } P(X \leq t) = \frac{p}{100}$$

$$\Rightarrow 1 - e^{-\lambda t} = \frac{p}{100}$$

$$\Rightarrow e^{-\lambda t} = 1 - \frac{p}{100} = \frac{100 - p}{100}$$

$$\Rightarrow e^{\lambda t} = \frac{100}{100 - p}$$

$$\lambda t = \ln \left(\frac{100}{100 - p} \right)$$

$$t = \frac{1}{\lambda} \ln \left(\frac{100}{100 - p} \right)$$

$$t = n \times \ln \left(\frac{100}{100 - p} \right)$$

Hence, option b is correct.

For example: n=18 and p=60

Given mean of exponential random variable (life of light bulb) = 18 months.

$$\Rightarrow \frac{1}{\lambda} = 18, \Rightarrow \lambda = \frac{1}{18}.$$

$$\text{Given } P(X \leq t) = 0.6$$

$$\Rightarrow 1 - e^{-\lambda t} = 0.6$$

$$\Rightarrow e^{-\lambda t} = 0.4$$

$$\lambda t = \ln 2.5$$

$$\frac{1}{18}t = \ln 2.5$$

$$t = 18 \ln 2.5$$